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Semi-classical methods, angular momentum, and non-analytic problems

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Abstract

We explore the semi-classical Sommerfeld method of quantization as applied to states with nonzero angular momentum, and show that it leads to qualitatively and quantitatively useful information about systems with spherically symmetric potentials. We begin by reviewing the traditional application of this model to hydrogen, and discuss the way Einstein–Brillouin–Keller (EBK) quantization resolves a mismatch between Sommerfeld states and true quantum mechanical states. We then analyze systems with logarithmic and Yukawa potentials, and compare the results of Sommerfeld and EBK quantization to those from solving Schrödinger’s equation. We show that the semi-classical quantization techniques provide insight into the spread of energy levels associated with a given principle quantum number, as well as giving quantitatively accurate approximations for the energies. We also argue that analyzing systems in this manner involves an interesting application of numerical methods, as well as providing insight into the connections between classical and quantum mechanical physics.

Keywords: quantum mechanics, old quantization, teaching numerical methods

 Supplementary material for this article is available [online](#)

(Some figures may appear in colour only in the online journal)

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1. Introduction

The origins of quantum mechanics are usually dated to 1905, with the publication of Einstein's work on the photoelectric effect [1], even though the study of Schrödinger's equation and matrix mechanics—what we usually think of as quantum mechanics—was primarily developed in the mid 1920s [2, 3]. During the intervening two decades, quantum mechanics consisted of a series of ad hoc techniques which combined classical reasoning with simple rules for quantization. Most of us today are familiar with one of the earliest pieces of this story: in 1911 Niels Bohr developed a model of the atom which places electrons in classical circular orbits but quantizes the allowed values of angular momentum, so as to generate a discrete energy spectrum [4]. This model, which accurately produces the energy spectrum of hydrogen in spite of involving a misleading physical picture, is usually included in introductory modern physics courses.

However, almost no undergraduate courses spend time on the extensions that followed from this initial work. Instead, they leap awkwardly to a discussion of wavefunctions and Schrödinger's equation, without making any real connection between the two ideas. Furthermore, these courses often end before reaching later approximation techniques such as the WKB method, which do create a link between these separate ideas. As a result, many students never develop a real understanding semi-classical approximations, in spite of the fact that they are effective at capturing a range of qualitative and quantitative information, and still play an active role in serious theoretical physics: consider for example the analyses of strings in curved spacetime backgrounds, where true quantization is not well understood [5].

In the period after Bohr's work was published, Arnold Sommerfeld, William Wilson, and Jun Ishiwara independently developed extensions of the Bohr model [6–8]. These models put the electron in a hydrogen atom into classical elliptical orbits, and then imposed two separate quantization conditions on integrals over radial and angular momentum. This 'Sommerfeld' model reproduces some aspects of degeneracy in the hydrogen spectrum: multiple different combinations of the two quantum numbers lead to the same energy levels, different energy levels do not have the correct number of states. To resolve that difficulty, it is necessary to consider both the Einstein–Brillouin–Keller (EBK) method [9–12] and the way a quantum mechanical energy spectrum is altered by changing the number of spatial dimensions. This reveals that the quantum numbers involved in the old quantization analysis should be shifted by half-integers, which does not alter the energy levels of hydrogen but does correct the degeneracy mismatch.

We present here an analysis of two other systems: confinement in a logarithmic potential, and in a Yukawa potential. These systems were studied in [13], with a focus on the zero-angular momentum states. Here, we extend this work to consider non-zero angular momentum, where the lack of degeneracy in the energy spectrum allows us to clearly see the mapping between semi-classical states and exact states. The analysis also demonstrates that some information about the spectra in these systems can be obtained through simple analytic calculations.

For both the logarithmic and Yukawa potentials, finding much of the semi-classical and Schrödinger energy spectra requires a variety of different numerical methods. As such, a project such as this one also provides an opportunity for educating students about these tools, which include root finders, numerical integrators, and differential equation solvers. Not only do students need to understand how to use these tools separately, they must also synthesize them in order to execute both the old quantization and Schrödinger's equation calculations. Finally, in implementing these methods students need to rely on qualitative and analytic analyses to inform the structure of the computations.

In section 2 of this paper, we review the basics of Sommerfeld quantization, and its application to modeling the hydrogen atom. We also discuss the mismatch between the number of states present at each energy level in old quantization and in an analysis of Schrödinger's

equation, and how this can be resolved by using EBK quantization. In section 3, we explore the comparison for a logarithmic potential. We show that analytically tractable aspects of semi-classical analysis give insight into the ‘spread’ of different energies associated with the same quantum number, and that a good quantitative agreement can be found when using the EBK semi-classical method. In section 4, we perform similar comparisons for the Yukawa potential. In addition to supporting the conclusions drawn from studying the logarithmic potential, we explore the existence of a finite number of bound states in this system. In section 5, we discuss some of the details of the numerical methods used and their pedagogical value, and in section 6, we summarize our results and suggest possible avenues of future work.

2. Hydrogen: a state counting problem

We will begin by reviewing Sommerfeld quantization in the traditional case of ‘hydrogen’ (that is, an attractive Coulombic potential). We will therefore assume we have a particle of mass m in three dimensions, subject to the spherically symmetric attractive potential

$$V_H(r) = -\frac{C}{r}. \quad (1)$$

Classically, angular momentum conservation guarantees that our particle remains confined to a single ‘orbital plane’, which means we can describe the motion using radial and angular coordinates $r(t)$ and $\theta(t)$. (We can also choose to orient the orbital plane such that the orbital motion is counterclockwise.) The coordinates $r(t)$ and $\theta(t)$ satisfy energy and angular momentum conservation equations

$$E = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + V(r) = \frac{p_r^2}{2m} + U_{\text{eff}}(r; p_\theta), \quad (2)$$

and

$$p_\theta = mr^2\dot{\theta}, \quad (3)$$

where E is the energy, and p_θ is the magnitude of the angular momentum vector. The object $p_r = m\dot{r}$ is the radial momentum, which is not conserved, and $U_{\text{eff}}(r; p_\theta)$ is the ‘effective potential’, which governs the radial motion. For the Coulomb potential, the bound orbits consistent with these equations are closed ellipses with one focal point at the origin.

In Sommerfeld quantization [6–8], we then impose quantization conditions on the action integrals over radial and angular momentum:

$$\hbar n_r = \oint p_r dr, \quad \hbar n_\theta = \oint p_\theta d\theta, \quad (4)$$

where we consider $n_r, n_\theta = 0, 1, 2, 3, \dots$, with $n_r + n_\theta > 0$. As originally proposed, these conditions are ‘ad hoc’ generalizations of the Bohr model. Each integral is performed over a complete cycle of the coordinate r or θ , respectively. Angular momentum conservation reduces the second condition to $p_\theta = \hbar n_\theta$. For the radial quantization integral, a complete cycle consists of the particle moving from minimum distance r_- to maximum distance r_+ and back again, allowing us to replace $\oint(\dots)dr$ with $2 \int_{r_-}^{r_+}(\dots)dr$. We can find these turning points by setting $p_r = 0$ in equation (2) and solving for r . Finally, we can use the energy conservation equation to rewrite p_r in terms of r , so that the radial quantization condition becomes

$$\hbar n_r = 2 \int_{r_-}^{r_+} \sqrt{2m[E - U_{\text{eff}}(r; \hbar n_\theta)]} dr, \quad (5)$$

which can be done analytically in the case of the Coulomb potential (remembering we will have $E < 0$). First we simplify the integral with the definitions $E_R = \frac{mC^2}{2\hbar^2}$ the Rydberg energy, $a = \frac{\hbar^2}{mC}$ the Bohr radius, and $\rho = \frac{r}{an_\theta^2}$. Then we complete the square, using $w = \frac{n_\theta^2|E|\rho}{E_R\varepsilon} - \frac{1}{\varepsilon}$, where $\varepsilon = \sqrt{1 - \frac{n_\theta^2|E|}{E_R}}$ is the eccentricity of the elliptical orbit. Then we make a trigonometric substitution $w = \cos \theta$. These steps give us

$$\begin{aligned} \frac{n_r}{n_\theta} &= \frac{1}{\pi} \sqrt{\frac{n_\theta^2|E|}{E_R}} \int_{\rho_-}^{\rho_+} \sqrt{(\rho - \rho_-)(\rho_+ - \rho)} \frac{d\rho}{\rho} \\ &= \frac{\varepsilon^2}{\pi} \sqrt{\frac{E_R}{n_\theta^2|E|}} \int_{-1}^1 \frac{\sqrt{1-w^2} dw}{1+\varepsilon w} = \frac{\varepsilon^2}{\pi} \sqrt{\frac{E_R}{n_\theta^2|E|}} \int_0^\pi \frac{\sin^2 \theta d\theta}{1+\varepsilon \cos \theta}. \end{aligned} \quad (6)$$

which is

$$\begin{aligned} \frac{n_r}{n_\theta} &= \frac{1}{\pi} \sqrt{\frac{E_R}{n_\theta^2|E|}} \int_0^\pi \left[1 - \varepsilon \cos \theta - \frac{(1-\varepsilon^2)}{1+\varepsilon \cos \theta} \right] d\theta \\ &= \sqrt{\frac{E_R}{n_\theta^2|E|}} - \frac{\sqrt{1-\varepsilon^2}}{\pi} \int_0^\pi \frac{d\theta}{1+\varepsilon \cos^2 \frac{\theta}{2} - \varepsilon \sin^2 \frac{\theta}{2}}. \end{aligned} \quad (7)$$

Finally, we can make the substitution $u = \sqrt{\frac{1-\varepsilon}{1+\varepsilon}} \tan \frac{\theta}{2}$, which allows us to write

$$\frac{n_r}{n_\theta} = \sqrt{\frac{E_R}{n_\theta^2|E|}} - \frac{1}{\pi} \int_0^\infty \frac{du}{1+u^2} = \sqrt{\frac{E_R}{n_\theta^2|E|}} - 1. \quad (8)$$

Solving for energy then gives us

$$\text{Sommerfeld Hydrogen Spectrum:} \quad E = -\frac{E_R}{(n_r + n_\theta)^2}. \quad (9)$$

Two special types of states in this analysis deserve particular consideration. If we choose $n_r = 0$, then we are restricting ourselves to circular orbits. The radial quantization integral disappears, and energy quantization is given by

$$E = U_{\text{eff}}(r_c; n_\theta), \quad U'_{\text{eff}}(r_c; \hbar n_\theta) = 0. \quad (10)$$

Here, r_c is the radius of a circular orbit, which can be found by minimizing the effective potential. (When we use the Coulomb potential in these equations, we recover equation (9), with $n_r = 0$.) These are the states used in the ‘Bohr model’ of an atom [4], and can also be thought of as the states which maximize angular momentum, for a given amount of energy.

On the other hand, if we choose $n_\theta = 0$ we are considering ‘orbits’ which consist entirely of radial motion. These are most easily visualized as a limit of a classical orbit in which we allow $p_\theta \rightarrow 0$, which leads to the radial quantization integral

$$\hbar n_r = 2 \int_0^{r_+} \sqrt{2m[E - V(r)]} dr, \quad (11)$$

with the upper limit of the integral determined by $E = V(r_+)$. Using the Coulomb potential in this integral again leads back to equation (9), this time with $n_\theta = 0$:

$$\begin{aligned} n_r &= \frac{\sqrt{2m|E|}}{\pi\hbar} \int_0^{\frac{C}{|E|}} \sqrt{\frac{C}{|E|r} - 1} \, dr \\ &= \frac{4}{\pi} \sqrt{\frac{E_R}{|E|}} \int_0^1 \sqrt{1 - u^2} \, du = \sqrt{\frac{E_R}{|E|}} \Rightarrow E = -\frac{E_R}{n_r^2}, \end{aligned} \quad (12)$$

where $u^2 = \frac{|E|r}{C}$, and the final integral is just the area of a quarter circle of radius 1. Sommerfeld himself considered these states unphysical and discounted them. However, we know now that the solutions to Schrödinger's equation do include states with zero angular momentum, represented by spherically symmetric wavefunctions.

Now consider the radial part of Schrödinger's equation, which is

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + U_{\text{eff}}(r; \tilde{p}_\theta) u(r) = Eu(r), \quad \frac{\tilde{p}_\theta^2}{\hbar^2} = \ell(\ell + 1) \quad (13)$$

where $R(r) = \frac{u(r)}{r}$ is the radial wavefunction, and ℓ is the quantum number associated with the magnitude of angular momentum (see for example [14, 15]). The solutions to this equation consistent with bound state boundary conditions $u(0) = u(\infty) = 0$ give us the famous result

$$\text{Schr. Hydrogen Spectrum:} \quad E = -\frac{E_R}{n^2}. \quad (14)$$

Here, $n = 1, 2, 3, \dots$ is the 'principle quantum number', and the angular momentum quantum number is constrained to $\ell = 0, 1, 2, \dots, n - 1$.

Clearly, in this case Sommerfeld quantization leads to the same energy levels as Schrödinger's equation, and just as clearly, we should identify the principle quantum number as $n = n_r + n_\theta$. However, if we try to compare the number of states at each energy level, we see a discrepancy. In the Schrödinger spectrum, the ground state is unique: we must choose $n = 1$ and $\ell = 0$. On the other hand, in the Sommerfeld argument there are two possibilities: $\{n_r = 0, n_\theta = 1\}$, and $\{n_r = 1, n_\theta = 0\}$. This pattern continues beyond the ground state level: at the n th level, there are n possible values for ℓ , but $n + 1$ possible values for n_θ . And since the energies of all of these states are the same, it is difficult to see from this perspective which Sommerfeld states should be mapped onto which wavefunction solutions to Schrödinger's equation.

The solution to this problem lies in EBK quantization [9–12], an approach which derives approximate (WKB) solutions to Schrödinger's equation, and then determines quantization conditions by requiring that the wavefunction be single-valued. The results come close to agreeing with the Sommerfeld conditions. However, the quantization conditions are modified by phase shifts which occur at boundaries. Specifically, this adds a shift in dependence of the radial quantum number of $\frac{\mu}{4} + \frac{b}{2}$, where μ is the number of classical turning points in the radial coordinate $r(t)$, and b is the number of hard-wall reflections. In the case of the Coulomb system (as well as the logarithmic and Yukawa systems we will consider next), we have $\mu = 2$ and $b = 0$.

Furthermore, consider that classical orbits are confined to a plane by angular momentum quantization, and the Sommerfeld treatment above is therefore blind to the number of spatial dimensions the system is actually in. On the other hand, in $D \geq 2$ dimensions the radial

wavefunction for a hyper-spherically symmetric potential satisfies a wave equation [16]

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + U_{\text{eff}}(r; \tilde{p}_\theta) R(r) = ER(r), \quad \frac{\tilde{p}_\theta^2}{\hbar^2} = \left(\ell + \frac{D}{2} - \frac{3}{2} \right) \left(\ell + \frac{D}{2} - \frac{1}{2} \right), \quad (15)$$

where $R(r) = r^{(1-D)/2} u(r)$ is the radial wavefunction: as a result the true energy spectrum certainly should depend on the value of D . Based on the above form, in order to make our Sommerfeld/EBK results match true quantum mechanical results in D dimensions, we should expect to have to shift the dependence on n_θ by $\frac{D-2}{2}$.

Putting these pieces together, in order to get a good mapping between semi-classical states and three-dimensional Schrödinger's equation states, we can consider EBK states given by $n_r, n_\theta = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$, with $n = n_r + n_\theta$. This throws out *both* the circular orbits *and* the radial motion states, but recovers the correct energy levels and degeneracy counting.

The degeneracy associated with ℓ (or n_θ) present in the Coulomb system obscures what is going on with the mapping between the semi-classical and Schrödinger states, and perhaps makes the need for the above argument less than compelling (after all, we recover the correct energy levels from the original Sommerfeld argument). Since this degeneracy is a special feature of the Coulomb potential—a result of the conservation of the Runge–Lenz vector—we should gain additional insight by considering other spherically symmetric potentials.

3. The logarithmic potential

Suppose we now consider the same particle of mass m in three dimensions, but this time bound in a potential of the form

$$V_{\log}(r) = \beta \ln \left(\frac{r}{r_0} \right), \quad (16)$$

where β is a constant which sets the energy scale, and we choose $r_0 = \frac{\hbar}{\sqrt{m\beta}}$ for convenience. (Shifting the value of r_0 simply adjusts the entire energy spectrum upward or downward.)

Let us begin by using naive Sommerfeld quantization (integer values for n_r and n_θ), and start with the circular orbits, specified by $n_r = 0$. In this case, a straightforward application of equation (10) to our logarithmic potential gives

$$0 = U'_{\text{eff}}(r_c; \hbar n_\theta) = -\frac{\hbar^2 n_\theta^2}{mr_c^3} + \frac{\beta}{r_c} \Rightarrow r_c = \frac{\hbar n_\theta}{\sqrt{m\beta}} \quad (17)$$

which when used in the energy conservation equation gives

$$\text{Sommerfeld log Spectrum,} \quad n_r = 0 \quad E = \beta \left(\frac{1}{2} + \ln n_\theta \right), \quad n_\theta = 1, 2, 3, \dots \quad (18)$$

Similarly, we can consider the $n_\theta = 0$ scenario. This problem is also analytically tractable, and following the argument in [13], we can use equation (11) with the logarithmic potential to get

$$n_r = \frac{\sqrt{2}}{\pi r_0} \int_0^{r_0 e^{E/\beta}} \sqrt{\frac{E}{\beta} - \ln \frac{r}{r_0}} dr = \frac{2\sqrt{2}}{\pi} e^{E/\beta} \int_0^\infty x^2 e^{-x^2} dx = \frac{1}{\sqrt{2\pi}} e^{E/\beta}, \quad (19)$$

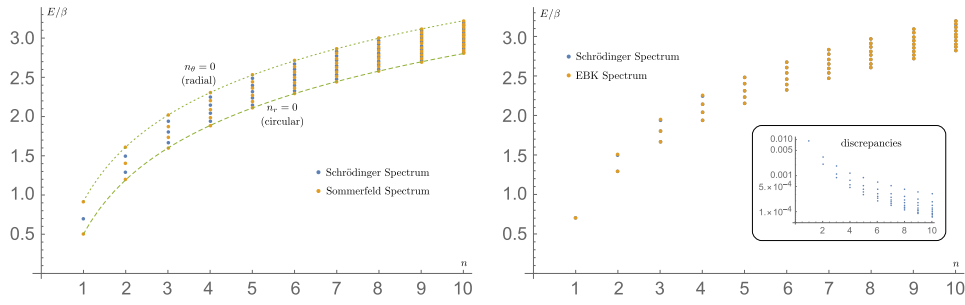


Figure 1. On the left, a plot showing the old Sommerfeld spectrum of the logarithmic potential, together with the Schrödinger spectrum. Lines indicating the parts of the Sommerfeld spectrum from radial motion and circular motion are also included. On the right, a plot showing the EBK spectrum together with the Schrödinger spectrum (the two spectra overlay each other almost exactly). The inset gives the discrepancies (absolute differences) between these values.

(where we are using the substitution $x = \sqrt{\frac{E}{\beta} - \ln \frac{r}{r_0}}$). This then gives

$$\text{Sommerfeld log Spectrum, } n_\theta = 0: \quad E = \beta \ln \left(n_r \sqrt{2\pi} \right), \quad n_r = 1, 2, 3, \dots \quad (20)$$

Apart from these special cases, finding the Sommerfeld spectrum requires numerical tools, as does solving Schrödinger's equation. (This is discussed more in section 5.) Figure 1 on the left shows a plot of the results of these analyses, in which we graph the energies vs the principle quantum numbers. For each value of n , there are multiple different energies, corresponding with different choices of ℓ in the exact spectrum and different choices of n_θ in the Sommerfeld spectrum.

We notice immediately that none of the Sommerfeld energies align well with the Schrödinger energies: a direct mapping between semi-classical states with integer values of $\{n_r, n_\theta\}$ and Schrödinger states with integer values of $\{n, \ell\}$ is not possible. However, we also see that for a given value of n , the energies of the Schrödinger spectrum always lie between the limiting Sommerfeld values associated with circular and radial motion. Furthermore, the larger n gets, the more our Schrödinger spectrum seems to 'fill in' the space between these extremes. As a result, just by using analytic Sommerfeld arguments, we gain insight into the 'spread' of energies associated with one principle quantum number. Notice in particular that this spread is actually a constant for the logarithmic system, whose value is $\Delta E = \frac{\beta}{2} (\ln(2\pi) - 1) \approx 0.4189\beta$.

Now consider EBK states, with $n_r, n_\theta = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$. In this case making the mapping $n_r + n_\theta = n$ and $\ell = n_\theta - \frac{1}{2}$ creates excellent agreement between the semi-classical states and the Schrödinger states, as is shown in figure 1 on the right, and in table 1. Note that the Schrödinger spectrum are consistent with values found previously [17], and the semi-classical analysis of this system is consistent with that performed in [18]. Of course, even with the EBK method we do not recover the true quantum mechanical energy spectrum. The agreement is only strong for large quantum numbers, and for more moderate values additional corrections are necessary [19].

Table 1. A table of the energies derived from Schrödinger's equation, as compared with EBK energies for the logarithmic potential, with the identifications $n_\theta = \ell + \frac{1}{2}$, and $n_r = n - \ell - \frac{1}{2}$. The numerical methods used to arrive at these values are described in section 5.

n	ℓ	$E_{\text{Schr.}}$	E_{EBK}	Discrepancies
1	0	0.697 759	0.706 894	0.009 135
2	1	1.294 57	1.296 59	0.002 02
	0	1.500 87	1.504 23	0.003 36
3	2	1.666 74	1.667 59	0.000 856
	1	1.804 37	1.805 51	0.001 133
	0	1.943 04	1.944 88	0.001 842
4	3	1.937 57	1.938 04	0.000 47
	2	2.040 86	2.041 43	0.000 57
	1	2.144 37	2.145 11	0.000 743
	0	2.249 13	2.250 33	0.001 199
5	4	2.150 54	2.150 84	0.000 296
	3	2.233 21	2.233 55	0.000 344
	2	2.315 92	2.316 33	0.000 413
	1	2.399 02	2.399 56	0.000 534
	0	2.483 36	2.484 21	0.000 86
6	5	2.326 06	2.326 26	0.000 203
	4	2.394 97	2.395 2	0.000 23
	3	2.463 87	2.464 14	0.000 265
	2	2.532 93	2.533 24	0.000 316
	1	2.602 43	2.602 84	0.000 407
	0	2.673 08	2.673 74	0.000 655

4. The Yukawa potential

We can also consider a potential of the form

$$V_Y(r) = -\frac{C}{r} e^{-r/R}, \quad (21)$$

where C and R are constants (once again with a particle of mass m in three dimensions). It is convenient to think of this a modification of the Coulomb potential (equation (1)), in which the exponential suppression factor introduces a rough ‘cut-off’ associated with the length scale R . This creates a system in which there will be only a finite number of bound states. For $r \ll R$, the exponential factor is insignificant, and as a result we expect to have some states which sit close to the origin and are approximately Coulombic. However, states sitting far away from the origin experience essentially zero force, and should therefore not be bound. Because the close-in states will have a size controlled by the ‘Bohr radius’ length scale $a = \frac{\hbar^2}{mC}$, the number of bound states will be determined by the ratio $\lambda = \frac{R}{a}$.

The inclusion of non-zero angular momentum further complicates this story. Consider the classical system, focussing on the effective potential shown in figure 2. For small values of p_θ , the system allows for bound states with negative energies. However, at a critical value of $p_\theta^* = \hbar\sqrt{2\lambda} e^{-1/2}$, these states disappear and we have $U_{\text{eff}} > 0$ for all r . At values of angular momentum just above this critical point, it is still possible to have classical bound states with positive energies (although tunneling should prevent the existence of quantum mechanical bound states in this regime). Finally, above a second critical value p_θ^{**} the effective potential

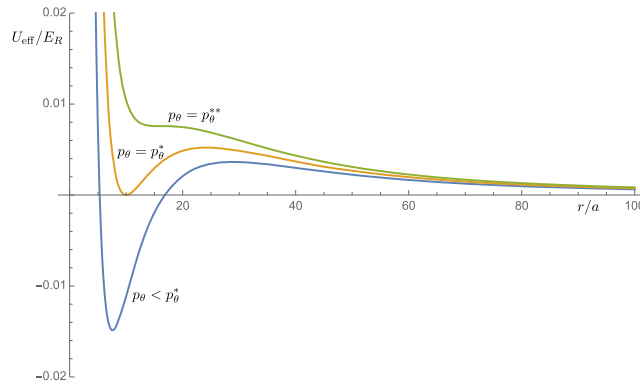


Figure 2. The effective potential for $\lambda = 10$, with choices $p_\theta < p_\theta^*$, $p_\theta = p_\theta^*$, and $p_\theta = p_\theta^{**}$.

ceases to have a minimum at all, so there are no longer any bound states in either the classical or the quantum mechanical system.

Now consider the Sommerfeld states corresponding to circular orbits, with $n_r = 0$. These states are found by minimizing the effective potential (equation (10)), which for the Yukawa potential leads to a transcendental equation:

$$0 = U'_{\text{eff}}(r_c; \hbar n_\theta) = -\frac{\hbar^2 n_\theta^2}{m r_c^3} + \frac{C}{r_c^2} e^{-r_c/R} + \frac{C}{r_c R} e^{-r_c/R}. \quad (22)$$

Nonetheless, we can identify a critical value for the quantum numbers n_θ :

$$n_\theta^* = \frac{p_\theta^*}{\hbar} = \sqrt{2\lambda} e^{-1/2}, \quad (23)$$

as the point where such a state can no longer have a negative energy, meaning we do not expect it to correspond to a true quantum mechanical bound state. Similarly, we can look for the Sommerfeld states corresponding to purely radial motion, with $n_\theta = 0$. These are found through equation (11), which again is not analytically tractable. However, following the arguments in [13], we can determine a critical value for the quantum numbers n_r :

$$n_r^* = 2\sqrt{\frac{\lambda}{\pi}}, \quad (24)$$

above which these states no longer have negative energy.

Figure 3 on the left shows the Sommerfeld spectrum and the Schrödinger spectrum, both obtained numerically. Notice that in this case the vertical axis is presented on a logarithmic scale, so as to better see the behaviors of the spectra when the energies are close to zero. This plot displays many of the same qualitative features discussed for the logarithmic case. Again, there is no clear mapping between semi-classical states corresponding to integer values of $\{n_r, n_\theta\}$, and Schrödinger states, and again the Schrödinger states always lie between the circular orbit Sommerfeld states and the radial motion Sommerfeld states. It is therefore still possible to think of the spread of possible quantized energies corresponding to a given choice of n as limited by these two extremes, though here the lack of analytic solutions makes this less useful.

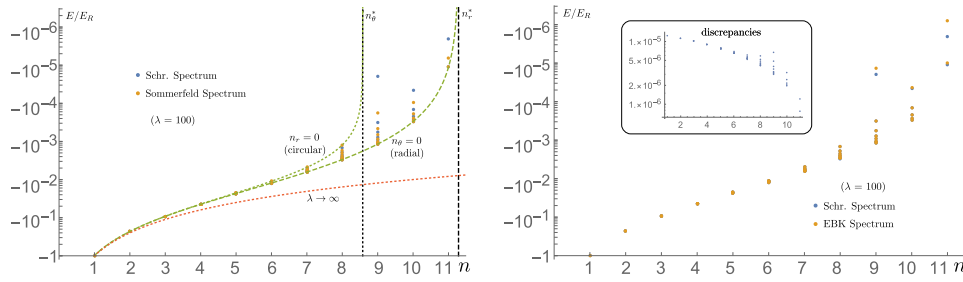


Figure 3. On the left, a plot showing the Sommerfeld spectrum of the Yukawa potential, together with the Schrödinger spectrum, for $\lambda = 100$. Additional lines indicate the parts of the Sommerfeld spectrum from radial motion and circular motion, the critical values n_r^* and n_θ^* , and the Coulomb spectrum. On the right, a plot showing the EBK spectrum together with the Schrödinger spectrum. The inset graph gives the discrepancies (absolute differences).

On the other hand, the two critical values n_r^* and n_θ^* do provide us with analytically accessible information that gives us insight into the Schrödinger system: we can see that for $n < n_\theta^*$, there are $n - 1$ bound states associated with principle quantum number n . However, once we are in the region $n_\theta^* < n < n_r^*$, there are fewer than $n - 1$ bound states for each n , and for $n > n_r^*$, there are no longer any bound states at all.

Finally, note that a line indicating the hydrogen spectrum (corresponding to $\lambda \rightarrow \infty$) is also included for reference. For small values of n , all of the Yukawa states lie close to the equivalent hydrogen states. As n becomes larger, the differences become more apparent: in particular that the Coulomb spectrum has an infinite number of negative energy bound states, while the Yukawa spectrum does not.

We can again obtain a much better agreement between the semi-classical states and the Schrödinger quantum mechanical states if we use EBK quantization, requiring $n_r, n_\theta = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$, with $n = n_r + n_\theta$ and $\ell = n_\theta - \frac{1}{2}$. Figure 3 on the right shows the Schrödinger spectrum with the EBK spectrum, and table 2 gives (some of) the numerical data. The EBK data here is consistent with that in [20], and the Schrödinger's data is consistent with that found in [21]. Note that the discrepancies generally become smaller as n increases, but the energies themselves do as well. Close to the limit where bound states cease to exist the discrepancies are comparable with the energies, so that the mapping begins to break down.

5. An exploration of numerical methods

In the previous two sections, we have argued that results obtained via semi-classical analysis can give qualitative insight into the true quantum mechanical spectrum (derived from Schrödinger's equation), as well as leading to quantitatively reasonable approximations once we take into account the half-integer shifts in EBK quantization. But a project such as this one can also serve an alternative educational purpose: as a chance for older undergraduate students to explore numerical methods. Most of the physics underlying this work is drawn from a standard undergraduate curriculum (classical orbits, energy eigenstates in Schrödinger's equation, and the Bohr model of the atom), and in the courses a typical upper division student has taken, they will have learned to apply this physics to analytically tractable problems. However, in order to execute the calculations for more generic potentials they must use a variety of basic numerical tools. Furthermore, they must assemble these tools in a sophisticated manner in order

Table 2. A table of some of the Schrödinger energies as compared with EBK energies for the Yukawa potential, for $\lambda = 100$, with the identifications $n_\theta = \ell + \frac{1}{2}$, and $n_r = n - \ell - \frac{1}{2}$. The numerical methods used to arrive at these values are described in section 5.

n	ℓ	$E_{\text{Schr.}}$	E_{EBK}	Discrepancies
1	0	−0.980 149	−0.980 137	0.000 0122
2	1	−0.230 491	−0.230 479	0.000 0114
	0	−0.230 587	−0.230 575	0.000 0114
3	2	−0.092 1229	−0.092 1126	0.000 0103
	1	−0.092 3062	−0.092 2959	0.000 0103
	0	−0.092 3977	−0.092 3874	0.000 0103
4	3	−0.044 1975	−0.044 1885	9.10×10^{-6}
	2	−0.044 4556	−0.044 4465	9.10×10^{-6}
	1	−0.044 6268	−0.044 6178	9.0×10^{-6}
	0	−0.044 7122	−0.044 7032	9.0×10^{-6}
5	4	−0.022 5323	−0.022 5244	7.90×10^{-6}
	3	−0.022 8508	−0.022 843	7.80×10^{-6}
	2	−0.023 0874	−0.023 0796	7.80×10^{-6}
	1	−0.023 2441	−0.023 2363	7.70×10^{-6}
	0	−0.023 3221	−0.023 3144	7.70×10^{-6}
6	5	−0.011 2818	−0.011 275	6.80×10^{-6}
	4	−0.011 6455	−0.011 6389	6.70×10^{-6}
	3	−0.011 9316	−0.011 925	6.60×10^{-6}
	2	−0.012 1433	−0.012 1368	6.50×10^{-6}
	1	−0.012 2832	−0.012 2768	6.50×10^{-6}
	0	−0.012 3528	−0.012 3464	6.40×10^{-6}

to actually obtain the data sets necessary for the comparison between OQ and Schrödinger energy states.

To complete this project, we used four basic numerical algorithms: a bisection routine for root finding, Simpson’s rule for numerical integration, the finite difference method to replace a differential equation with a matrix equation, and Runge–Kutta-4 for solving an initial-value-specified differential equation. For a good introduction to these techniques and many others, see [22]. However, one could in general replace these with other algorithms, or with built-in functions, to suit the audience. The full *Mathematica* code for completing this project is included in the supplementary material (<https://stacks.iop.org/EJP/42/035402/mmedia>).

Below, we present outlines of the numerical approaches as applied to the logarithmic potential (a very similar approach was used for the Yukawa potential). Note that two different techniques were used to find the eigenvalues of Schrödinger’s equation. We began by using the ‘finite difference method’, which approximates the Hamiltonian operator as a large, sparse matrix, thus allowing one to use linear algebra algorithms to find eigenvalues (we used pre-written code for the last part.) We also used the ‘shooting method’, in which an initial-value-driven differential equation solver (RK4, in our case) is combined with a root finder, and the energy eigenvalue is adjusted until the wavefunction satisfies appropriate boundary conditions. These techniques provide comparable numerical results, but have different pedagogical strengths: the shooting method allows one to focus on the connection between the spatial localization of a bound state (as expressed in boundary conditions) and the quantization of energy. On the other hand, the finite difference method creates a link between the ‘wavefunction’ approach to quantum mechanics which students generally learn first, and the ‘matrix mechanics’ interpretation often introduced in an advanced quantum mechanics course.

To assess numerical accuracy, we used the rough approach of modifying the numerical parameters controlling accuracy (doubling or halving them), and assessing how much this affected our answers. In all cases, we ensured that numerical errors would be small compared to the discrepancies between semi-classical and Schrödinger results.

5.1. Outline of numerical method to find semi-classical energies for the log potential

Our numerical approach works with dimensionless variables $\epsilon = E/\beta$ and $\rho = \frac{r}{r_0}$, so that we are attempting to find energies ϵ for each choice of $\{n_r, n_\theta\}$ satisfying

$$n_r = \frac{1}{\pi} \int_{\rho_-}^{\rho_+} \sqrt{2\epsilon - \frac{n_\theta^2}{\rho^2} - 2 \ln \rho} d\rho, \quad 0 = F(\rho_\pm; \epsilon, n_\theta) = 2\epsilon - \frac{n_\theta^2}{\rho_\pm^2} - 2 \ln \rho_\pm. \quad (25)$$

- (a) We created a function which takes inputs ϵ and n_θ and a numeric parameter ε , and outputs the limits of integration $\rho_\pm$ using a bisection routine (with ε determining the final bracket size). In order to do so, we need an ‘initial bracketing’ for each root:

- (1) Since $\rho = n_\theta$ is the minimum of the effective potential $V_{\text{eff}}(\rho) = \frac{n_\theta^2}{2\rho^2} - \ln \rho$, we should have $\rho_- \in [\varepsilon, n_\theta]$.
- (2) Furthermore, the term $-\frac{n_\theta^2}{\rho^2}$ can only decrease the value of the second root, so we should have $\rho_+ \in [n_\theta, e^\epsilon]$.

- (b) Next, we created a function which takes as inputs ϵ and n_θ , and numeric parameters ε and N , and outputs the integral given in equation (25). It calls the previous function to find the limits of integration, and then implements Simpson’s rule with N subdivisions.
- (c) Next, we created a function which takes as inputs both n_r and n_θ , as well as the numeric parameters ε and N , and an initial energy bracketing $\epsilon_{\min}$ and $\epsilon_{\max}$, and outputs the energy ϵ associated with those quantum numbers by again using a bisection routine.
- (d) Finally, we created ‘data collector’ function which takes as inputs the maximum desired values of n_θ and n_r , the numeric parameters ε and N , and a rough energy step size $d\epsilon$. Looping through values of both n_θ and n_r , it increases energy in steps $d\epsilon$ from minimum value $\epsilon_{\min} = \frac{1}{2} + \ln n_\theta$ until it finds an energy bracket for the desired value of n_r , calls the previous function to locate the eigenvalue ϵ , and stores this value in a list to be output at the end.

5.2. Outline of finite difference method to find ‘true’ QM energies for the log potential

Working with the same dimensionless variables, we now want to find the eigenvalues ϵ of the equation

$$-\frac{1}{2} \frac{d^2 u}{d\rho^2} + \left[\frac{\ell(\ell+1)}{2\rho^2} + \ln \rho - \kappa \right] u = \tilde{\epsilon} u, \quad \epsilon = \tilde{\epsilon} + \kappa. \quad (26)$$

Our basic method is to use the finite difference method to generate a matrix equation approximation to the above, and find its eigenvalues using built-in linear algebra code. This code is most efficient at finding the eigenvalue with the smallest magnitude first, so we will be using equation (26) where κ is an approximation to the eigenvalue we are looking for.

- (a) First, we created a function which takes inputs ℓ , κ , and numeric grid parameters ρ_∞ and N (controlling the location of ‘numerical infinity’ and the number of interior grid points).

It then uses a basic sparse array matrix creator to generate the appropriate matrix, and then finds the eigenvalue $\tilde{\epsilon}$ of this matrix with the smallest magnitude, and returns the associated value ϵ .

- (b) Finally, we created a ‘data collector’ function which takes inputs ρ_∞ , N , and the list of eigenvalues and associated n and ℓ values found using the semi-classical method. It loops through this list, and uses each eigenvalue as the approximation κ input for the previous function, generating the ‘true’ eigenvalue, and storing that in an output list.

5.3. Outline of shooting method to find ‘true’ QM energies for the log potential

This method is also solving equation (26), but with $\kappa = 0$. We will be using Runge–Kutta-4, which is a method for solving a first order differential equation, so we begin by rewriting equation (26) as

$$\vec{v}(\rho) = \begin{bmatrix} u(\rho) \\ u'(\rho) \end{bmatrix}, \quad \frac{d\vec{v}}{d\rho} = \mathbb{M}\vec{v}, \quad \mathbb{M} = \begin{bmatrix} 0 & 1 \\ \frac{\ell(\ell+1)}{\rho^2} + 2 \ln \rho - 2\epsilon & 0 \end{bmatrix} \quad (27)$$

for which we will use initial conditions $\vec{v}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. We want to find values of ϵ for which $u(\rho_\infty) = 0$ (the value of $u(\rho)$ at some ‘numerical infinity’).

- (a) First, we created a function which takes as inputs ℓ , the numeric parameters ρ_∞ and N , and the energy ϵ , and uses an RK4 algorithm to approximate the value of $u(\rho_\infty)$ by taking N steps.
- (b) Then, we created a function which takes as inputs ℓ , the numeric parameters ρ_∞ , N , and ϵ , and an initial energy bracketing $\epsilon_{\max}$ and $\epsilon_{\min}$, and uses the previous function inside a bisection routine to find the energy eigenvalue ϵ .
- (c) Finally, we created a ‘data collector’ function. As with the finite difference approach, we ‘seed’ this with estimates of the energies generated by the semi-classical method. Looping through this list, for each eigenvalue it uses a small bracket around that value, and inputs this in the previous function, to generate the eigenvalue ϵ and store that in an output list.

6. Conclusions

The semi-classical techniques developed by Sommerfeld, Wilson, and Ishiwara, along with the more accurate EBK method, are generally left out of an undergraduate education in quantum mechanics. In most cases, courses jump straight from the Bohr model of the atom to a discussion of Schrödinger’s equation, and frequently end before reaching the point of discussing the WKB approximation. This is unfortunate, since semi-classical methods can often lead to both qualitative and quantitative insights, and in fact are still useful in current research. In this project, we explored the connection between semi-classical analysis and Schrödinger analysis through the lens of the logarithmic and Yukawa potentials, by using a series of numerical techniques.

We began by reviewing the structure of Sommerfeld quantization in the context of the Coulomb potential. This allowed us to see that while the Sommerfeld argument produces the correct energy levels and is consistent with the existence of degeneracy, it does not make it clear how precisely to map semi-classical states onto Schrödinger states. To solve the problem, it is necessary to consider the more correct EBK method, which shows us that we need

to shift the quantum numbers appearing in the semi-classical analysis by half-integers. In order to explore this further, we considered the logarithmic and Yukawa potentials. There, we noted that the ‘limiting cases’ of Sommerfeld analysis—corresponding to circular and radial motion—effectively bracket the states given by Schrödinger’s equation. Thus, Sommerfeld analysis can provide insight into the spread of energy states associated to loss of degeneracy; this insight was particularly significant for the logarithmic potential, where the limiting cases are approachable by analytic calculations. For the Yukawa potential, analytic tools were only sufficient to make predictions about the points where states begin to become unbound. We also showed that by implementing the half-integer shifts in quantum numbers required by EBK quantization, a clear mapping between semi-classical and Schrödinger states could be established, such that the semi-classical results would provide quantitatively accurate approximations.

Although the ability of semi-classical analysis to provide quick, pen-and-paper approximations to problems that are otherwise not analytically tractable is often emphasized, in this case both the semi-classical energy states and the Schrödinger states were generally provided by numerical means. In fact, the range of numerical methods used and the complexity of assembling them makes this type of project useful as a way to introduce computational physics to older students, while simultaneously giving them greater insight into quantum mechanics. As a result, we also briefly laid some of these numerical details in section 5 and discussed their various pedagogical values.

It would be interesting to extend this project to consider the states in the Yukawa potential which are classically bound, but with positive energies. These states exist at the same energy and angular momentum as scattering states, and quantum mechanics allows for tunneling between the two. In the true quantum mechanical system, these should therefore not correspond to bound states, but instead to resonances, and a mapping might be created between the semi-classical states and these resonances. One could also extend this project in a straightforward way by choosing other potentials to explore; for example a potential might be analyzed in which tunneling would exist between multiple bound states, instead of between a bound state and a scattering state. Alternatively, potentials directly connected to physical systems might be considered, and comparison with experimental data could be incorporated into the analysis.

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